Vibe Shuffle: Next-Song Selection from Relative Valence–Arousal Trajectories

Markus Baumann*
Ludwig-Maximilians-Universität München
Munich, Germany
markus.baumann@campus.lmu.de

Felix Hajuj
Ludwig-Maximilians-Universität München
Munich, Germany
felix.hajuj@campus.lmu.de

Miriam Janner
Ludwig-Maximilians-Universität München
Munich, Germany
M.Janner@campus.lmu.de

Abstract

Music recommenders are good at learning what listeners generally like, but the song that fits can depend on how they feel in the moment. *Vibe Shuffle* investigates whether camera and cardiac changes relative to personal references can support more context-sensitive next-song recommendations. The system records personal reference values for the available sensing channels and combines camera-based facial and movement cues with heart rate and heart-rate variability. During playback, changes from those references form a short trajectory in a two-dimensional valence–arousal space without assigning the listener a fixed emotion label. The trajectory mean identifies a catalog region, and the nearest unplayed song in that region is selected. An exploratory within-session crossover across 15 sessions compared Vibe Shuffle with Random selection from the same catalog. Mood-fit ratings were higher on average under Vibe Shuffle, while general liking changed little, but both estimates were too uncertain to establish a recommendation benefit. Vibe Shuffle connects changes in camera and cardiac signals to explicit next-song decisions without claiming to identify the listener's emotion.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Information systems** → *Recommender systems.*

Keywords

affective computing, music recommendation, relative affect, valence and arousal, heart-rate variability, facial cues, personal references

1 Introduction

Music recommendation is usually based on what a person tends to like. Listening history, favorite artists, similar users, and song features all provide useful signals of long-term preference. Yet the song that fits can change from one moment to the next. A track that works during exercise may feel out of place during a quiet break, and a song that someone likes in general may still not be the song they want to hear next. Because people use music to maintain, change, and regulate mood and activation [11, 25], this difference between general liking and momentary fit is an important challenge for contextual music recommendation [12, 26].

One way to include the current situation is to ask listeners to select their mood. This can provide useful context, but it also interrupts the listening experience. Automatic adaptation avoids that interruption, although the available observations are uncertain. Facial behavior varies with pose, lighting, culture, context, and deliberate expression [2, 5, 14]. Cardiac timing is also affected by respiration, posture, movement, fitness, medication, and sensor quality [15, 23, 27]. Neither channel should therefore be treated as a direct measurement of emotion.

Vibe Shuffle examines whether short-term changes within the same listener provide useful context for selecting the next song. At the beginning of a session, the system records personal reference values for the available camera and cardiac channels. Later observations are expressed only as changes from these references and mapped to relative valence and arousal coordinates. The observations collected during a song form a trajectory. Its mean identifies one of four regions in a fixed music catalog, and the nearest unplayed track in that region becomes the next recommendation. The region names Energetic, Calm, Tense, and Melancholic describe parts of the catalog rather than the listener's emotional state. Physiological arousal and musical energy are likewise used as matching coordinates without being treated as the same construct.

Vibe Shuffle combines three elements:

- **Sensing relative to personal references.** Optional camera and cardiac observations are combined without assigning a fixed emotion label.
- **Explicit next-song selection.** A trajectory mean, catalog region, and nearest-track rule turn those observations into one decision.
- **Exploratory comparison.** A within-session crossover across 15 sessions compared Vibe Shuffle with Random selection from the same catalog and measured momentary mood fit separately from general liking.

Mood-fit ratings were higher on average under Vibe Shuffle, whereas liking changed little. However, the uncertainty around both differences included zero, so the crossover did not establish improved recommendation quality. In a future music service, the method could re-rank tracks supplied by a conventional preference model. Evaluating that use will require validated signal mappings, complete trial-level records, and a preregistered study with a realistic candidate pool.

2 Background

Dimensional affect models provide a simple vocabulary for listeners and music. Russell's circumplex places valence from unpleasant to pleasant and arousal from low to high activation [22, 24]. Thayer's

*Corresponding author.

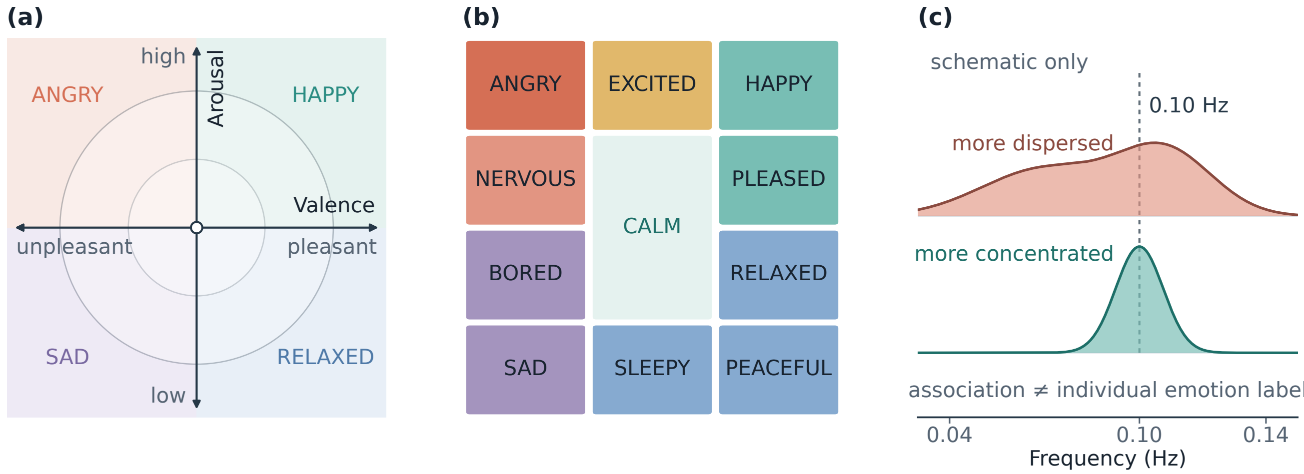

Figure 1: Panels (a)–(b) adapt the affect-space representations of Helmholz et al. [9], based on Russell's circumplex [24] and Thayer's mood–arousal model [28]. Panel (c) schematically represents the group-level frequency distinction reported by Balaji et al. [4].

model adds categories such as energetic and tense [28]. These models are common in music-emotion research, but their meaning depends on the labels, stimuli, and context [7]. Vibe Shuffle uses them as a practical coordinate system for recommendation, not as a complete description of experience.

The continuous and categorical representations in Figure 1 define the recommendation space, while the HRV panel provides frequency context. Vibe Shuffle outputs neither emotion labels nor coherence profiles. Its coordinates encode changes from personal reference values rather than direct measurements of emotion.

Songs are represented in a parallel valence–energy space. The catalog valence coordinate describes the positivity conveyed by a track, while energy captures its intensity and activity. Catalog valence and energy have served as matching axes for affective valence and arousal [9]. Musical energy and physiological arousal remain distinct constructs, and music-emotion coordinates are subjective and may change during a song [1, 30].

Splitting both axes at 0.5 creates four catalog regions. The mean coordinate of the listening trajectory determines the target region, and the remaining tracks in that region are ranked by their distance from this mean.

Cardiac timing provides one source of relative arousal evidence. HRV describes variation between heartbeats, while RMSSD summarizes short-term variation in RR intervals [27]. Their interpretation depends on recording length, respiration, movement, posture, fitness, medication, sensor quality, and device processing [15, 23]. Vibe Shuffle therefore uses heart rate and RMSSD only relative to a personal reference.

3 Related Work

Music can evoke, express, maintain, or regulate affect [11]. Emotion-aware recommenders therefore use the listener's current context to guide song selection. *Feel the Moosic* asks listeners to choose a position in an affect space and maps it to Spotify valence and energy ranges [9]. Vibe Shuffle instead derives its coordinate from camera and cardiac observations measured relative to personal references.

Other systems infer context from sensor data. Janssen et al. learned listener-specific relations between music and physiological responses and used them to guide an affective music player toward a target state [10]. Ayata et al. estimated valence and arousal from wearable measurements before recommending music [3]. Tran et al. matched facial estimates with nearby tracks in Spotify feature space [29]. Explicit reports, wearable measurements, and facial estimates can each provide recommendation context, but evidence that the resulting choices improve momentary fit remains limited.

Because facial and cardiac observations are uncertain and influenced by many factors, their role in song selection should remain understandable. Explanations can support transparency and trust [21], while perceived quality also depends on effort, privacy, and context [13].

Vibe Shuffle centers multimodal observations on personal reference values, aggregates their changes as a trajectory, and applies a fixed region-and-distance rule.

4 Concept Overview

Vibe Shuffle expresses camera and cardiac observations as changes from personal reference values rather than population thresholds. These relative observations form a trajectory mean, which determines a catalog region and selected track. Sensor outputs guide selection but do not identify the listener's emotion.

The selection objective is momentary track fit, not mood regulation. The method does not assume that an energetic track should reduce activation or that positive valence should counteract a negative state. Maintaining, intensifying, and shifting affect would require separate adaptation rules.

The method combines personal reference values, a multi-sample trajectory, a fixed region-and-distance rule, and a Random comparator using the same catalog. Missing sensing channels do not stop a session, and the active signal path is recorded.

Vibe Shuffle follows a six-step loop: acquire personal references, collect camera and cardiac observations during playback, express them as relative valence–arousal points, summarize the trajectory, choose the nearest eligible song in the corresponding catalog region, and record the rating and decision fields. Adaptation occurs between tracks, so brief fluctuations cannot interrupt the current song (Figure 2).

These design choices serve different purposes. Personal references make the system sensitive to changes within one listener rather than differences between people. Updating only between tracks keeps playback stable. The four catalog regions provide an understandable target, while distance ranking selects the closest remaining song within that target. Optional sensing lets the same pipeline operate with camera data, cardiac data, both, or a fallback coordinate. The result is a direct and inspectable chain from observation to song choice.

5 Signal and Pipeline

The multimodal input has two distinct channels. A camera supplies visual and behavioral cues; an optional wearable supplies physiological timing. The camera reference runs for up to 14 seconds and may be skipped. The cardiac reference, when available, records 120 seconds of device-produced heart rate and RR intervals. Subsequent observations are centered on these personal values. The references do not define a neutral or true emotional state; they provide local comparison points for the session.

Camera observations primarily provide relative valence evidence, whereas cardiac changes provide relative activation evidence. Both mappings require independent validation.

Once the references are established, each channel contributes to the relative coordinate.

MediaPipe Face Landmarker runs locally and returns facial blendshape activations [17]. A fixed heuristic combines selected smile, cheek, brow, eye, jaw, frown, and mouth cues into positive, negative, and tension-related scores. Samples scheduled at 120 ms intervals are debiased against the reference and smoothed across frames. More positive visible evidence moves relative valence right; negative or tension-related evidence moves it left. A slowly adapting reference follows longer changes. If no face is detected, valence returns to the center.

Visible movement and nodding can also add positive-valence and upward-arousal evidence. The channel represents change in observable behavior under a fixed rule, not felt emotion. Lighting, pose, occlusion, morphology, culture, deliberate expression, dancing, and repositioning may change the output [2, 5, 14]. Fixed coefficients and gating rules are applied uniformly across sessions.

Cardiac evidence enters through a compatible Bluetooth Heart Rate Service device, which supplies beats per minute and, when available, RR intervals. This interface exposes neither raw electrocardiography nor the device's beat-detection logic. Implausible or abruptly changing intervals are rejected before variability is calculated. The reference stores median heart rate, the root mean square of successive differences (RMSSD), and robust spread estimates.

During listening, higher-than-reference heart rate and lower-than-reference RMSSD provide upward arousal evidence; changes in the opposite direction provide downward evidence. Heart rate receives the larger influence, and RMSSD is damped when the indicators disagree. At least 20 accepted RR intervals are required for the full estimate. RMSSD is a recognized short-term variability summary, but duration, respiration, posture, movement, fitness, medication, contact quality, and vendor processing affect its interpretation [15, 23, 27]. Agreement between 120-second and longer RMSSD measurements has been reported under controlled resting conditions, not short listening windows with movement [19].

SDNN and a custom spectral-peakedness measure termed physiology coherence are computed for diagnostic purposes only. This measure differs from the group-level coherence measure associated with post-session emotion labels in large-scale biofeedback data [4] and does not affect song selection.

The channel-specific values enter a shared pipeline for acquisition, plausibility filtering, centering on personal references, smoothing, quality assessment, and coordinate fusion. Usable cardiac evidence sets the arousal base, while camera movement can add upward evidence. Camera movement supplies an upward-only fallback when cardiac data are unavailable. If neither channel is usable, the coordinate returns to the central reference point.

Camera frames, facial landmarks, and Bluetooth notifications are processed in browser memory. Stored study data comprise derived summaries, reference status, signal source, quality flags, timing, selected-song metadata, and ratings. Frames, landmarks, raw packets, waveforms, complete trajectories, personal facial-reference values, and the session seed are not stored.

6 Inference and Adaptation

During confirmed playback, an event-driven update to the fused state appends a point to the trial trajectory

$$\mathcal{T} = \{(V_1, A_1), \ldots, (V_m, A_m)\}.$$

At rating time, the arithmetic mean

$$(\bar{V}, \bar{A}) = \frac{1}{m} \sum_{i=1}^{m} (V_i, A_i)$$

becomes the target for the next Vibe choice. Aggregation reduces dependence on one observation but is event-weighted: periods that generate more fused-state updates contribute more points than periods with fewer updates. Sample count and listening duration are recorded for each trial.

Splitting both axes at 0.5 produces four catalog regions: Energetic, Calm, Tense, and Melancholic. These names identify valence–energy partitions; they do not classify the listener. Listener arousal and song energy are corresponding axes used for matching, not equivalent constructs [9, 30].

The frozen catalog contains 100 unique Spotify tracks, with 25 in each region. Track valence and energy coordinates were obtained from a public historical dataset [18].

Selection applies the same exclusion rule in both conditions: the current track and every previously played track are removed. Let

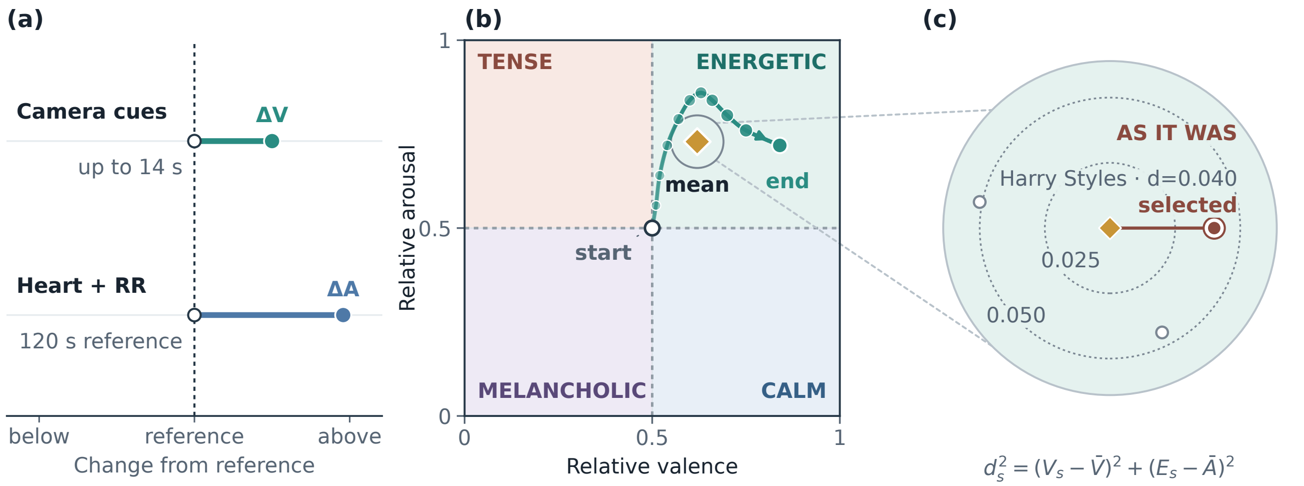

Figure 2: Panel (a) shows signed camera and cardiac changes measured against personal references. It shows the primary signal paths; camera movement may also contribute to relative arousal. Panel (b) summarizes the trajectory between its open start and filled endpoint using the arithmetic mean (gold diamond). In panel (c), the nearest catalog track is *As It Was* by Harry Styles, which the selection rule therefore selects. Its Euclidean distance is shown beside the track; two unlabeled open points indicate nearby alternatives, and the equivalent squared-distance expression appears below the panel. Panels (a)–(b) are schematic, whereas panel (c) uses catalog-derived coordinates and distance. Coherence is logged but not used for selection.

$Q(\bar{V}, \bar{A})$ denote the set of remaining tracks in the catalog region containing the mean trajectory coordinate. Vibe Shuffle selects the smallest Euclidean distance:

$$s^* = \arg \min_{s \in Q(\bar{V}, \bar{A})} \sqrt{(V_s - \bar{V})^2 + (E_s - \bar{A})^2}.$$

Equal-distance candidates are ordered by a deterministic score derived from the track identifier and session seed. If the region is exhausted, the method searches the complete remaining catalog. Both conditions use the same catalog, exclusion rules, number of tracks, playback interface, and rating questions. Vibe Shuffle selects the nearest remaining track in the target region, whereas Random ranks all remaining tracks by a deterministic pseudorandom score derived from the track identifier and a session-derived seed.

Because sensing is optional, a Vibe decision can use camera evidence, cardiac evidence, both, or neither. The first Random track is selected pseudorandomly, whereas a first-block Vibe track uses the fused coordinate available at session start. At a block transition, a first Vibe track uses the final listening window of the preceding block. Later Vibe tracks use the mean trajectory from the immediately preceding window.

Adaptation changes only the next-song ranking. At selection time, the method computes a target region, eligible set, distances, and selected track. Recorded selection variables include reference status, target coordinate, trajectory sample count, selected-track distance, region match, selected track, and quality indicators. Complete trajectories, eligible sets, and full distance lists are not stored.

7 User Study and Results

7.1 Study Design

The study examined whether context-sensitive selection improves the momentary fit of the next songs while maintaining general liking. Fifteen sessions followed a within-session crossover design. Each session contained a five-track Vibe block and a five-track Random block drawn from the same frozen catalog. The block order was reversed across two study sequences, and condition labels were hidden in the interface. Each session therefore provided a paired comparison between the two selection methods.

Each track played for up to 60 seconds, with the option to continue to the rating screen earlier. After every track, listeners answered *How much do you like this song?* and *How well did it fit your current mood?* on seven-point scales. They also selected one of five descriptive mood categories: Energetic, Calm, Tense, Melancholic, or Neutral. This categorical response did not affect song selection and was not included in the analyses.

In the Vibe condition, the next song was chosen from the relative valence–arousal trajectory described in Sections 5 and 6. In the Random condition, selection did not use that trajectory. Both conditions used the same catalog, number of tracks, playback interface, and rating questions. The selection method was the central experimental difference.

7.2 Analysis

The analysis compared the overall experience of the two five-track blocks. Ratings were averaged within each session and condition, and the resulting condition means served as the unit of analysis. Paired differences were defined as Vibe minus Random.

For mood fit and liking, Table 1 reports condition means and standard deviations, paired mean differences with two-sided 95% confidence intervals, and standardized paired effects d_z. Directional paired t tests and Wilcoxon signed-rank tests assess whether ratings were higher under Vibe. The confidence intervals use the paired t distribution with 14 degrees of freedom. Figure 3 additionally shows 20,000 fixed-seed bootstrap resamples of the paired mean distribution; its displayed interval bars remain t-based. The analysis was exploratory and not preregistered, so estimation and two-sided uncertainty are primary [6, 20].

7.3 Results

Mood fit averaged 4.71 under Vibe Shuffle and 4.15 under Random selection. The paired difference was +0.56 rating points, with a 95% confidence interval from −0.21 to 1.33 and a standardized paired effect of $d_z = 0.40$. Eight session means were higher under Vibe, one was equal, and six were lower. The directional paired test yielded $t(14) = 1.56$, $p = .070$, and the signed-rank test yielded $W^+ = 74.0$, $p = .088$.

General liking averaged 4.56 under Vibe and 4.47 under Random. The paired difference was +0.09 points, with a 95% confidence interval from −0.52 to 0.71 and $d_z = 0.08$. The directional tests yielded $t(14) = 0.32$, $p = .375$, and $W^+ = 58.5$, $p = .353$. The mood-fit point estimate was larger than the liking estimate, but the two effects were not directly compared. Both confidence intervals include zero, so their size and direction remain uncertain.

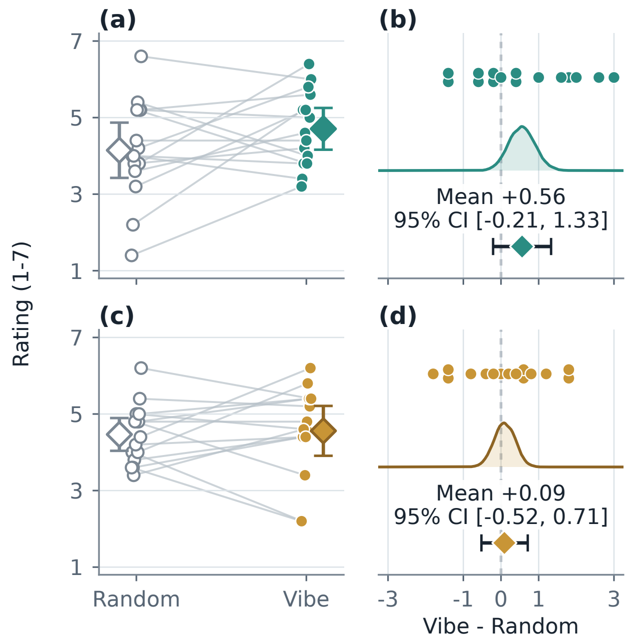

Figure 3: Mood fit is shown in panels (a)–(b) and liking in panels (c)–(d). The left panels connect the Random and Vibe five-track means for each session; diamonds and bars summarize the condition means and their t-based 95% confidence intervals. The right panels show all paired differences, fixed-seed bootstrap distributions of the mean difference, and t-based 95% confidence intervals. Positive differences favor Vibe Shuffle.

8 Discussion, Limitations, and Future Work

8.1 Discussion

The two outcomes capture different aspects of recommendation quality. Liking reflects general preference, whereas mood fit asks whether a song suits the present situation. A context-aware method can therefore target momentary fit without aiming to increase general liking. The outcomes were analyzed separately; neither estimate was precise enough to support a performance conclusion.

The selection rule links camera and cardiac changes to a recommendation through four explicit steps: personal references, a relative trajectory, a catalog region, and a distance ranking. This sequence keeps the recommendation logic understandable without assigning the listener an emotion label. Mood-fit differences nevertheless varied across sessions. The available averages cannot determine whether this variation arose from signal availability, trajectory shape, familiarity, order, or the songs remaining in the catalog.

Vibe Shuffle is intended to complement rather than replace preference modeling. A conventional recommender could first identify songs that the listener is likely to enjoy, after which the relative trajectory could re-rank that candidate set for the current session. Whether this combination improves momentary fit remains a hypothesis for future evaluation.

Vibe Shuffle also supports a lightweight deployment model. Camera frames, facial landmarks, and Bluetooth notifications are processed in browser memory, while stored data are limited to derived summaries, reference and quality status, timing, song metadata, and ratings. Raw camera and cardiac streams are therefore not retained. Since camera and wearable inputs are optional, a session can continue with either channel, both channels, or a fallback coordinate. These properties make the method suitable as a re-ranking component in an existing music service.

Sixty-three deterministic unit tests cover catalog structure, data serialization, sensing heuristics, and song ranking. They check the implemented decision rules; they do not validate sensor accuracy or recommendation benefit.

8.2 Limitations

The fixed catalog and Random condition improve experimental control but do not reproduce candidate generation or preference modeling in a full music service. The two conditions could also select different songs and present them in a different order. The measured difference therefore reflects the complete recommendation experience produced by each method.

The crossover controls stable differences within a session, although practice, fatigue, and carryover may still affect the second block [8]. Condition labels were hidden in the interface, but study administration could still reveal them. Fifteen sessions leave broad confidence intervals and do not support stable order, subgroup, or hardware estimates. The small average liking difference does not establish that liking is preserved because no equivalence or non-inferiority range was defined. Five-track condition means can also hide differences between individual songs.

Table 1: Five-track condition means across 15 sessions. Differences are Vibe minus Random. Confidence intervals are two-sided; p-values correspond to one-sided Vibe-over-Random tests.

Outcome	Vibe M (SD)	Random M (SD)	Difference [95% CI]	$t(14)$, p	W^+, p	d_z
Mood fit	4.71 (0.99)	4.15 (1.30)	+0.56 [−0.21, 1.33]	1.56, .070	74.0, .088	0.40
Liking	4.56 (1.18)	4.47 (0.77)	+0.09 [−0.52, 0.71]	0.32, .375	58.5, .353	0.08

The sensing layer uses fixed mappings from facial behavior, movement, heart rate, and RMSSD to relative valence and arousal. These observations can change with pose, lighting, respiration, posture, physical movement, fitness, medication, and sensor quality [2, 5, 14, 15, 23, 27]. Movement may influence camera evidence and cardiac measurements simultaneously, and the study did not estimate the contribution of each sensing channel. The trajectory mean is event-weighted, because it is computed over event-driven fused-state updates rather than time-weighted samples. The four-region split also introduces hard boundaries at 0.5. These choices make the decision rule simple, but they are not the only ways to summarize and match a trajectory. The study did not separately measure sensor accuracy or playback failures, so their influence on the ratings is unknown.

8.3 Future Work

The sensing layer should be evaluated separately from song selection. Brief within-listener valence and arousal reports collected after selected trials could be compared with the system coordinates. Camera-only, cardiac-only, combined, and no-signal variants would then show which channels add useful information. These reports would assess the mapping rather than enter the recommendation rule.

The trajectory summary should also be studied directly. Event-weighted averaging can be compared with time-weighted averaging, a recent-window estimate, and robust summaries that reduce the influence of isolated changes. The region-first rule can be compared with continuous nearest-track matching in the full valence–energy space. These comparisons would separate the effects of sensing, temporal aggregation, and catalog matching.

A preference-based recommender provides the practical comparator. A standard model could generate a small candidate set of songs that the listener is likely to enjoy, and Vibe Shuffle could re-rank the same candidates using the current trajectory. Holding the candidate set fixed would isolate the value of session context from that of general preference modeling.

A larger preregistered crossover [16, 20] should use mood fit as the primary outcome and liking as a safeguard. Sample size should be based on the smallest mood-fit improvement that would matter in practice. A predefined equivalence or non-inferiority margin for liking would test whether contextual re-ranking improves momentary fit without making the selected songs less enjoyable. Track-level analyses could examine how effects develop across a session and differ between sensing variants. Effort, privacy, and willingness to enable a camera or wearable should be measured alongside recommendation quality [13].

9 Conclusion

Vibe Shuffle provides an explicit method for context-sensitive next-song selection. Camera and cardiac observations are measured relative to personal references, combined into a short valence–arousal trajectory, and translated into a catalog region and nearest eligible track. The system uses observable changes as recommendation context without presenting them as direct measurements of emotion.

Across 15 sessions, mood fit differed by +0.56 points and liking by +0.09 points, but both confidence intervals included zero. The available data therefore do not establish either effect. A larger study is needed to determine whether the method improves momentary fit while preserving general liking.

Vibe Shuffle links observations measured relative to personal references to a next-song choice through a trajectory mean and a fixed region-and-distance rule. Validating the signal mappings and testing this rule as a re-ranking layer above a standard preference model are the next steps. This positions short-term, within-listener context as an inspectable complement to preference modeling, not a substitute for it.

10 Contribution Statement

Markus Baumann, Felix Hajuj, and Miriam Janner jointly developed the project concept, conducted the study, interpreted the results, and reviewed the manuscript. Markus coordinated the implementation, analysis, figures, and preparation of the final paper.

Open Science and Transparency

To support transparency and reproducibility, the complete project repository is openly available at https://github.com/eybmits/vibe-shuffle. It includes the source code, catalog data, automated tests, figure-generation scripts, manuscript sources, rendered paper, and project website. Together, these materials allow others to inspect the system's decision logic, reproduce the software and paper outputs, and extend Vibe Shuffle in future research.

References

[1] Anna Aljanaki, Yi-Hsuan Yang, and Mohammad Soleymani. 2017. Developing a Benchmark for Emotional Analysis of Music. *PLOS ONE* 12, 3 (2017), e0173392. doi:10.1371/journal.pone.0173392

[2] Hillel Aviezer, Yaacov Trope, and Alexander Todorov. 2012. Body Cues, Not Facial Expressions, Discriminate Between Intense Positive and Negative Emotions. *Science* 338, 6111 (2012), 1225–1229. doi:10.1126/science.1224313

[3] Değer Ayata, Yusuf Yaslan, and Mustafa E. Kamasak. 2018. Emotion Based Music Recommendation System Using Wearable Physiological Sensors. *IEEE Transactions on Consumer Electronics* 64, 2 (2018), 196–203. doi:10.1109/TCE.2018.2844736

[4] Sai Balaji, Nachum Plonka, Mike Atkinson, Malathy Muthu, Minvydas Ragulskis, Alfonsas Vainoras, and Rollin McCraty. 2025. Heart Rate Variability Biofeedback in a Global Study of the Most Common Coherence Frequencies and the Impact of Emotional States. *Scientific Reports* 15, 1 (2025), 3241. doi:10.1038/s41598-025-87729-7

[5] Lisa Feldman Barrett, Ralph Adolphs, Stacy Marsella, Aleix M. Martinez, and Seth D. Pollak. 2019. Emotional Expressions Reconsidered: Challenges to Inferring Emotion From Human Facial Movements. *Psychological Science in the Public Interest* 20, 1 (2019), 1–68. doi:10.1177/1529100619832930

[6] Geoff Cumming. 2014. The New Statistics: Why and How. *Psychological Science* 25, 1 (2014), 7–29. doi:10.1177/0956797613504966

[7] Tuomas Eerola and Jonna K. Vuoskoski. 2013. A Review of Music and Emotion Studies: Approaches, Emotion Models, and Stimuli. *Music Perception* 30, 3 (2013), 307–340. doi:10.1525/mp.2012.30.3.307

[8] Anthony G. Greenwald. 1976. Within-Subjects Designs: To Use or Not to Use? *Psychological Bulletin* 83, 2 (1976), 314–320. doi:10.1037/0033-2909.83.2.314

[9] Patrick Helmholz, Michael Meyer, and Susanne Robra-Bissantz. 2019. Feel the Moosic: Emotion-Based Music Selection and Recommendation. In *Proceedings of the 32nd Bled eConference: Humanizing Technology for a Sustainable Society*. University of Maribor Press, Maribor, Slovenia, 203–221. doi:10.18690/978-961-286-280-0.11

[10] Joris H. Janssen, Egon L. van den Broek, and Joyce H. D. M. Westerink. 2012. Tune in to Your Emotions: A Robust Personalized Affective Music Player. *User Modeling and User-Adapted Interaction* 22 (2012), 255–279. doi:10.1007/s11257-011-9107-7

[11] Patrik N. Juslin and Daniel Västfjäll. 2008. Emotional Responses to Music: The Need to Consider Underlying Mechanisms. *Behavioral and Brain Sciences* 31, 5 (2008), 559–575. doi:10.1017/S0140525X08005293

[12] Marius Kaminskas and Francesco Ricci. 2012. Contextual Music Information Retrieval and Recommendation: State of the Art and Challenges. *Computer Science Review* 6, 2–3 (2012), 89–119. doi:10.1016/j.cosrev.2012.04.002

[13] Bart P. Knijnenburg, Martijn C. Willemsen, Zeno Gantner, Hakan Soncu, and Chris Newell. 2012. Explaining the User Experience of Recommender Systems. *User Modeling and User-Adapted Interaction* 22, 4–5 (2012), 441–504. doi:10.1007/s11257-011-9118-4

[14] Peter Kuppens, Francis Tuerlinckx, Michelle Yik, Peter Koval, Joachim Coosemans, Kevin J. Zeng, and James A. Russell. 2017. The Relation Between Valence and Arousal in Subjective Experience Varies With Personality and Culture. *Journal of Personality* 85, 4 (2017), 530–542. doi:10.1111/jopy.12258

[15] Sylvain Laborde, Emma Mosley, and Julian F. Thayer. 2017. Heart Rate Variability and Cardiac Vagal Tone in Psychophysiological Research—Recommendations for Experiment Planning, Data Analysis, and Data Reporting. *Frontiers in Psychology* 8 (2017), 213. doi:10.3389/fpsyg.2017.00213

[16] Daniël Lakens. 2022. Sample Size Justification. *Collabra: Psychology* 8, 1 (2022), 33267. doi:10.1525/collabra.33267

[17] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, Wan-Teh Chang, Wei Hua, Manfred Georg, and Matthias Grundmann. 2019. MediaPipe: A Framework for Building Perception Pipelines. doi:10.48550/arXiv.1906.08172 arXiv:1906.08172 [cs.DC]

[18] maharshipandya. 2022. Spotify Tracks Dataset. Kaggle. https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset.

[19] M. Loretto Muñoz, Arie van Roon, Harriëtte Riese, Chris Thio, Emma Oostenbroek, Iris Westrik, Eco J. C. de Geus, Ron Gansevoort, Joop Lefrandt, Ilja M. Nolte, and Harold Snieder. 2015. Validity of (Ultra-)Short Recordings for Heart Rate Variability Measurements. *PLOS ONE* 10, 9 (2015), e0138921. doi:10.1371/journal.pone.0138921

[20] Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. 2018. The Preregistration Revolution. *Proceedings of the National Academy of Sciences* 115, 11 (2018), 2600–2606. doi:10.1073/pnas.1708274114

[21] Ingrid Nunes and Dietmar Jannach. 2017. A Systematic Review and Taxonomy of Explanations in Decision Support and Recommender Systems. *User Modeling and User-Adapted Interaction* 27, 3–5 (2017), 393–444. doi:10.1007/s11257-017-9195-0

[22] Jonathan Posner, James A. Russell, and Bradley S. Peterson. 2005. The Circumplex Model of Affect: An Integrative Approach to Affective Neuroscience, Cognitive Development, and Psychopathology. *Development and Psychopathology* 17, 3 (2005), 715–734. doi:10.1017/S0954579405050340

[23] Karen S. Quigley, Peter J. Gianaros, Greg J. Norman, J. Richard Jennings, Gary G. Berntson, and Eco J. C. de Geus. 2024. Publication Guidelines for Human Heart Rate and Heart Rate Variability Studies in Psychophysiology—Part 1: Physiological Underpinnings and Foundations of Measurement. *Psychophysiology* 61, 9 (2024), e14604. doi:10.1111/psyp.14604

[24] James A. Russell. 1980. A Circumplex Model of Affect. *Journal of Personality and Social Psychology* 39, 6 (1980), 1161–1178. doi:10.1037/h0077714

[25] Thomas Schäfer, Peter Sedlmeier, Christine Städtler, and David Huron. 2013. The Psychological Functions of Music Listening. *Frontiers in Psychology* 4 (2013), 511. doi:10.3389/fpsyg.2013.00511

[26] Markus Schedl, Hamed Zamani, Ching-Wei Chen, Yashar Deldjoo, and Mehdi Elahi. 2018. Current Challenges and Visions in Music Recommender Systems Research. *International Journal of Multimedia Information Retrieval* 7 (2018), 95–116. doi:10.1007/s13735-018-0154-2

[27] Fred Shaffer and J. P. Ginsberg. 2017. An Overview of Heart Rate Variability Metrics and Norms. *Frontiers in Public Health* 5 (2017), 258. doi:10.3389/fpubh.2017.00258

[28] Robert E. Thayer. 1991. *The Biopsychology of Mood and Arousal*. Oxford University Press, New York, NY.

[29] Hieu Tran, Tuan Le, Anh Do, Tram Vu, Steven Bogaerts, and Brian Howard. 2023. Emotion-Aware Music Recommendation. *Proceedings of the AAAI Conference on Artificial Intelligence* 37, 13 (2023), 16087–16095. doi:10.1609/aaai.v37i13.26911

[30] Yi-Hsuan Yang and Homer H. Chen. 2012. Machine Recognition of Music Emotion: A Review. *ACM Transactions on Intelligent Systems and Technology* 3, 3, Article 40 (2012), 30 pages. doi:10.1145/2168752.2168754